

Why the Deterministic #Knapsack Record Stands

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Abstract

The fastest deterministic FPTAS for #Knapsack is the $\tilde{O}(n^{2.5}\varepsilon^{-1.5})$ algorithm of Gawrychowski–Markin–Weimann [1], unimproved since 2018. This memo, written to be read alongside [1], explains why: the algorithm’s single expensive primitive – the pairwise merge of two sparsified sum-approximation lists – is, as a computational task, equivalent in the worst case to general $(\min, +)$ -convolution. Under the standard MinConv hardness hypothesis, the merge is optimal, and with it every FPTAS in the sum-approximation framework. The reduction’s core is machine-checked in Lean. We state precisely what an improvement would now require, and what remains possible beyond the worst case.¹

1 The question

[1] computes a $(1 + \varepsilon)$ -sum approximation of the knapsack counting function by divide and conquer: split the items, recurse, and merge the two children’s sparsified lists by convolution (their Section 4). Every recursion level pays the same $\tilde{O}(n^{2.5}\varepsilon^{-1.5})$, and all of it is one primitive: a merge of two s -point lists costs s^2 pairwise products, almost all immediately discarded by re-sparsification. Any merge in s^{2-c} time would improve the record polynomially. Can the merge be faster?

2 The answer

Theorem 1 (conditional optimality of the merge). *Computing the value-sparsified, position-exact merged frontier of two s -point sum-approximation lists requires $s^{2-o(1)}$ time in the worst case, unless the MinConv hypothesis fails.*

Three ingredients, two of them published:

1. **The ramp (machine-checked).** Adding $R \cdot k$ to any bounded integer sequence makes it monotone while shifting every anti-diagonal of its $(\min, +)$ -convolution uniformly – so monotone MinConv with unbounded values is exactly general MinConv. Sparsified break-point sequences are monotone with unbounded positions; their merge cannot lean on the fast bounded-monotone MinConv algorithms (most recently the deterministic $n^{1.5+o(1)}$ of Jin–Park–Saha–Xu [3]).

¹The Lean development accompanying this memo verifies the reduction end to end - encoding, executable merge cumulative, sandwich, read-off, padding, and ramp, composed in one statement - against the same verified code that runs the FPTAS; see the repository.

2. **The read-off (machine-checked).** Encode a ramped MinConv instance as two breakpoint lists with unit value-steps: the merged frontier’s inverse at index-sum m is the MinConv value at m . Value-only sparsification slack does not blur the read-off: the positions reported are exact. So a fast merge is a fast general MinConv.
3. **The floor (published).** Exact Pareto sum – the same task in different clothing – has no $O(n^{2-\delta})$ algorithm under the MinConv hypothesis, even when the output has linear size [2].

3 Why the escape routes close

Rounding positions instead. Allowing capacity-side error breaks the merge barrier (that is the “weak approximation” world, where bounded-monotone MinConv applies and $\tilde{O}(n + \varepsilon^{-1.5})$ -type bounds exist [5, 3]). But an FPTAS for #Knapsack is a *strong* approximation: the capacity is exact. Any scheme that quantizes positions must, at the root, resolve the mass within a capacity window around C – and isolating that window is NP-hard (its innermost case is counting exact Subset-Sum solutions, whose nonzeroness is Subset-Sum itself). Cumulative counting from zero, with exact positions, is forced.

Exploiting additive structure of the input. 3SUM remains hard on Sidon sets [4]: low additive energy of positions is not, by itself, an algorithmic lever. Structure helps only when it is promised (bounded range, monotone values) – which the ramp shows the merge does not enjoy.

Scope, stated plainly. The theorem is conditional (MinConv hypothesis) and framework-level: it binds every algorithm that computes sum-approximation merges – which includes all known deterministic FPTASes for #Knapsack (Gopalan et al., Štefankovič et al., Rizzi–Tomescu, Halman, and [1]) – but it is not an unconditional lower bound for the problem. An improvement now requires either refuting the MinConv hypothesis or a deterministic framework that avoids position-exact merges entirely while surviving the window-isolation hardness. Both are recognized-hard open problems.

4 Beyond the worst case

The obstruction is sharply localized: merges are expensive exactly on additively structured inputs. Empirically (asserted probes in the repository), a curvature dichotomy holds: windows of the breakpoint sequence are either near-affine – compressible by relaxed piecewise-linear fits, mergeable in closed form – or curved, in which case near-minimal ties on anti-diagonals are rare and a pruned sweep is cheap. An adaptive merge built on this dichotomy is subquadratic on every structured family we tested, and is blocked only on the MinConv-hard core, as Theorem 1 says it must be. Making this a beyond-worst-case theorem – and porting the resolution-lever ideas from the randomized side where sampling is permitted – is the live continuation.

References

- [1] P. Gawrychowski, L. Markin, O. Weimann. *A Faster FPTAS for #Knapsack*. ICALP 2018.
- [2] S. Funke, D. Hespe, P. Sanders, S. Storandt, C. Truschel. *Pareto Sums of Pareto Sets*. ESA 2023.

- [3] C. Jin, J. Park, B. Saha, Y. Xu. *Deterministic Monotone Min-Plus Product and Convolution*. ICALP 2026. arXiv:2605.07150.
- [4] C. Jin, Y. Xu. *Removing Additive Structure in 3SUM-Based Reductions*. STOC 2023.
- [5] K. Bringmann, A. Cassis. *Faster Knapsack Algorithms via Bounded Monotone Min-Plus Convolution*. ICALP 2022.